Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/06/2007
PI Name:	Hammond, Timothy M.D.		
Project Title:	Transcription factors mediating gene expression changes during renal cell rotating wall vessel culture.		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed end date to 9/30/2006 per P. Denkins/JSC (8/07)		
Key Personnel Changes/Previous PI:	Hammond, Timothy		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1344		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Although there are several precedents for the transduction of mechanical forces into changes in gene expression, including heat shock proteins and shear stress responses, the molecular mechanisms by which mechanical culture conditions in the rotating wall vessel modulate gene expression in cultured cells remain largely unknown. Several characteristics of renal cells made they especially well suited for rotating wall vessel studies. First, reformation of microvilli, and expression of clinically important proteins such as P450 drug metabolism enzymes, and drug and hormone receptors provides a unique toxicology tool not available in any other form. Second, the in situ milieu of renal epithelial cells has shear levels similar to those in the rotating wall vessel. Last, several lines of evidence suggest defined transcription factors (from the vitamin D receptor, and TFII-I to TFII-B) translocated to the nucleus in a time dependent manner during initiation of rotating wall vessel culture. This application aims to utilize renal cells as a model system to identify and validate molecular mechanisms by which selected transcription factors control changes in specific clusters of genes during rotating wall vessel culture. Unifying hypothesis Nuclear translocation of specific transcription factors mediates changes in expression of specific gene clusters during rotating wall vessel culture of renal cells. Specific Aim I: To identify kinetic gene clusters responding to the “simulated microgravity” of rotating wall vessel mechanical culture conditions. Specific Aim II: To test whether blocking translocation of individual transcription factors specificity reverses the effects rotating wall vessel culture on select gene clusters. This aim will focus on the effects of up and down regulation of nuclear translocation of the vitamin D receptor nuclear translocation on gene expression clusters (Specific Aim IIa), as well as identifying other transcription factors for future studies (Specific Aim IIb). If successful, our work using micro-array technology will demonstrate specific clusters of genes, which change expression during the rotating wall vessel form of suspension culture, and some of the transcription factors, which mediate the changes. This will elucidate specific molecular mechanisms critical for optimization of suspension culture. This facilitates the production of bioproducts such as hormones from the kidney and other tissues, as well as new methods for testing drug interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has two practical application to life on earth. The first application is to understand hoe to make cheap, clean bioproducts for use as biopharmaceuticals. This work is also designed to facilitate a quantum leap in studies of chemosensitivity by allowing individual drug regimens to be designed based on in vitro analysis of a patient diseased or normal tissue.
Task Progress:	This renal epithelial cell data has been compared for organ specificity using data sets from human liver cells flown in space by Stelsys LLC, to the whole organ kidney and liver effects by comparison to tissue gene expression in mice flown on STS-108 by Amgen, and mechnistically by comparison to a simple eukaryotic system in the yeast Saccharomyces cerevisiae. This has allowed comparison of organ specific changes, determination how good ground-based models are as predictors of microgravity effects, and prediction of transcription mediated drug metabolism changes in whole animals and astronauts in space. This study explores the connection between changes in gene expression and the genes that determine strain survival during suspension culture, using the model eukaryotic organism, Saccharomyces cerevisiae. The Saccharomyces cerevisiae homozygous diploid deletion pool (HDDP), and the BY4743 parental strain were grown for 18 h in a rotating wall vessel (RWV), a suspension culture device optimized to minimize the delivered shear. In addition to the reduced shear conditions, the RWVs were also placed in a static position or in a shaker in order to change the amount of shear stress on the cells. Using simple linear regression, it was found that there were 140 differentially expressed genes for which >70% of the variation can be explained by shear stress alone. A significant number of these genes are involved in catalytic activity. In the HDDP, shear stress was associated with significant survival changes in 15 deletion strains (R(2>) > 0.7) Interestingly, both analyses uncovered changes in the ribosomal protein machinery. Comparing the changes in gene expression and strain survival under the different shear conditions allows for the insights into the molecular mechanisms behind the cells response to shear stress. This in turn can provide information for the optimization of suspension culture.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Johanson K, Allen PL, Gonzalez-Villalobos RA, Nesbit J, Nickerson CA, Höner zu Bentrup K, Wilson JW, Ramamurthy R, D’Elia R, Muse KE, Hammond J, Freeman J, Stodieck LS, Hammond TG. "Haploid deletion strains of Saccharomyces cerevisiae that determine survival during space flight." Acta Astronaut. 2007 Feb-Apr;60(4-7):460-71. http://dx.doi.org/10.1016/j.actaastro.2006.09.011 , Feb-2007
Articles in Peer-reviewed Journals	Coleman CB, Gonzalez-Villalobos RA, Allen PL, Johanson K, Guevorkian K, Valles JM, Hammond TG. "Diamagnetic levitation changes growth, cell cycle, and gene expression of Saccharomyces cerevisiae." Biotechnol Bioeng. 2007 Nov 1;98(4):854-63. PubMed PMID: 17546692 [NOTE reported originally in Aug 2007 as Epub ahead of print] , Nov-2007
Articles in Peer-reviewed Journals	Johanson K, Allen PL, Gonzalez-Villalobos RA, Baker CB, D'Elia R, Hammond TG. "Gene expression and survival changes in Saccharomyces cerevisiae during suspension culture." Biotechnol Bioeng. 2006 Apr 20;93(6):1050-9. PMID: 16440349 , Apr-2006
Articles in Peer-reviewed Journals	Hammond DK, Elliott TF, Holubec K, Baker TL, Allen PL, Hammond TG, Love JE. "Proteomic retrieval from nucleic acid depleted space-flown human cells." Gravit Space Biol. 2006 Aug;19(2):145-6. http://gravitationalandspacebiology.org/index.php/journal/article/view/27 , Aug-2006